

VINOTHKUMAR P

Data Scientist | Data Analyst

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SUMMARY

Fresher Data Scientist and AI Developer skilled in Python, Machine Learning (regression, classification, clustering), SQL, and Power BI, with working knowledge of Pandas, NumPy, scikit-learn, and exposure to TensorFlow/PyTorch. Built an ensemble fraud detection model achieving 99.8% accuracy and a production-ready RAG-based AI chatbot using LLM and Streamlit. Experienced in end-to-end data pipelines: cleaning, EDA, feature engineering, model development, experimentation, documentation, and dashboard reporting, collaborating with cross-functional teams.

TECHNICAL SKILLS

Programming Languages: Python

Frontend: HTML, CSS

Libraries : Pandas, NumPy, Matplotlib, Seaborn, scikit-learn

Machine Learning: Supervised & Unsupervised Learning, Classification, Regression, Clustering, Ensemble Methods

Deep Learning : TensorFlow, PyTorch

Databases: MySQL, Oracle SQL (DDL, DML, Joins, Subqueries, Group By), PL/SQL

Visualization & BI: Power BI (DAX, KPIs, Slicers, Filters), Excel

Tools: Jupyter Notebook, VS Code, Google Colab, Databricks, Cursor, Codex, Groq API, Git/GitHub, Google Antigravity

PROJECTS

Credit Card Fraud Detection — Ensemble ML Classifier

Mar2026

- Built a stacked ensemble model (Random Forest + XGBoost + Gradient Boosting) to detect fraud on a highly imbalanced credit card dataset (~0.17% fraud rate).
- Applied SMOTE for class imbalance, conducted deep EDA, and tuned hyperparameters using GridSearchCV to maximize recall — minimizing missed fraud cases.
- Achieved 99.8% accuracy, 0.84 F1-Score, and 88% recall across stratified k-fold cross-validation; documented experiments and results for reproducibility.

RAG-Based Company Policy Chatbot — LLM Application

Apr2026

- Developed a Retrieval-Augmented Generation chatbot enabling employees to query internal HR policy PDF documents in natural language with zero hallucination.
- Built full RAG pipeline: PDF parsing, recursive text chunking, semantic retrieval, and Groq LLM response with strict prompt engineering for context-grounded answers only.
- Delivered interactive Streamlit UI with real-time document upload and conversation history, reducing HR policy query resolution time significantly.

INTERNSHIP

Data Science — PySpiders (Bangalore)

Oct2025-Apr2026

- Learned Python fundamentals (List, Set, Tuple, Dictionary) and OOPs concepts; used Pandas, NumPy, Matplotlib, and Seaborn for data cleaning and analysis.
- Built and evaluated ML models (regression, classification, clustering, ensemble algorithms) using Accuracy, F1-Score, Precision, Recall, R2, and MAE.
- Designed Power BI dashboards with DAX, KPIs, and interactive slicers to communicate data insights to cross-functional, non-technical stakeholders.
- Practiced Oracle SQL — DDL, DML, Joins, Subqueries, Group By, and Aggregations; documented experiments, processes, and model results for reproducibility.

EDUCATION

Master of Computer Applications (MCA)

Sep2024-May2026

Excel Engineering College, Namakkal, Tamil Nadu

CERTIFICATION

Data Science Certification — PySpiders (Bangalore)

Oct 2025-Apr2026

Python | SQL | Machine Learning | Power BI | Excel

ADDITIONAL

Languages: English, Tamil